Dissertation

Table of contents

[TODO: create cover, add boilerplate stuff]

Acknowledgements

[TODO: write; may be added after uploading dissertation to MyPhD]

Table of contents

[TODO: fix which sections are included]

Samenvatting

[TODO: write; may be added after uploading dissertation to MyPhD]

Glossary

[TODO: complete]

- complete
- imputed
- missing
- observed
- synthetic

Introduction

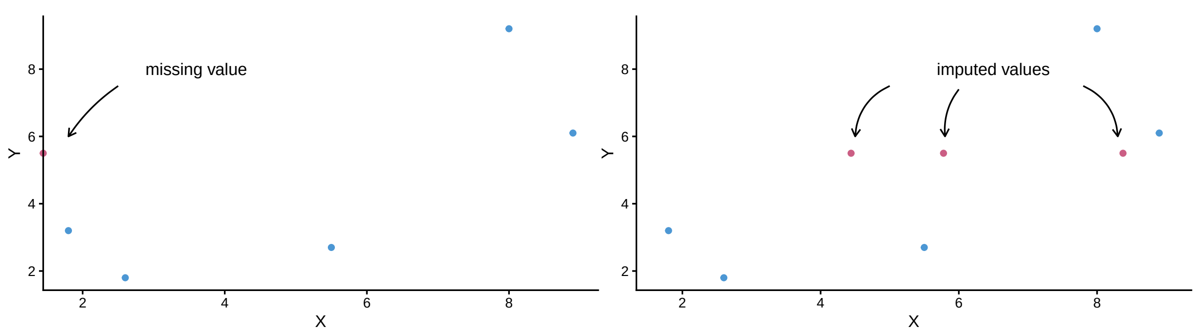
TL;DR (or: the quality of the solution)

This dissertation is about algorithms and the quality of their output. I investigate how a specific type of data-generating algorithm can be evaluated for use in statistical inferences. The type algorithm in question is the imputation algorithm. Imputation algorithms are algorithms with the purpose of generating plausible values based on a model of the observed information. This type of algorithm is popular for solving missing data problems.

Imputation algorithms are designed to 'impute' (i.e., fill in) missing values in incomplete datasets. Imputation is a popular and flexible procedure that separates the missing data problem from the analysis of scientific interest. Imputation allows data analysts to first solve for the missingness, to then be able to analyze the completed data as if they were completely observed.

The practice of drawing inference by replacing missing values with several synthetic substitutes is called 'multiple imputation'. Figure 1 provides an example of multiple imputation, where a missing value is replaced several times with plausible substitutes. The variability between imputations will reflect the uncertainty due to non-response in the final estimate.

Figure 1: Imputation of a missing value



Although imputation is widely supported with software, and many implementations of the method have convenient defaults, generating *good* imputations is not a trivial task. Evaluating whether these imputations should be trusted remains challenging.

How can we know when to trust the output imputation algorithms?

- Theoretical properties: analytically (often impossible)
- Empirical performance: via simulation studies ('in silico'; the standard, but not standardized)

- In practice/applied use: evaluate output, sensitivity analyses, ...? (very ad hoc, not systematic)

In this dissertation, I set out to investigate what is needed to develop, apply, and evaluate data-generating algorithms. I thereby focus on the imputation procedure 'MICE' as implemented in the R package `mice`.

Why? Because current evaluation practices (simulation design, convergence diagnostics, visual diagnostics) are fragmented and insufficient.

How? By bridging inferential evaluation and data-/algorithm-level diagnostics for imputation methods.

How did we get here? (or: statistical inference is just a tool)

My academic tag line has long been "statistician, interdisciplinarian, open scientist". And while there's no doubt I am an interdisciplinarian and an open scientist, I am no longer sure that I am a statistician necessarily. I am more inclined to call myself a methodologist rather than statistician.

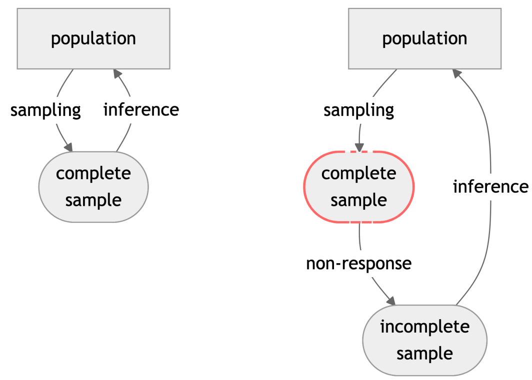
As a methodologist, I am interested in how science is done, and how it can be done better. Especially in the social and biomedical sciences, research should not be purely quantitative. Statistics is just *one* way of doing science. Don't get me wrong, statistics is still a favorite of mine. I love stats (I even have a mug that says so). Statistics is the science of uncertainty and extracting information from data (Hand, 2010). Extracting information from data and turn them into insights can feel magical. Statistical inference allows us to distinguish between real-world effects and random noise (Rosenberg & Curtain, 2015). I am most fan of the uncertainty of it all. I am motivated to contribute to trust in science by making it easier to reflect uncertainty in data analysis efforts, inferences, and graphs. In this dissertation, I hope to achieve this for missing data methodology.

Missing data problems are timeless, but technological advances have made treating missingness easier than ever. A single line of code can seem to solve your missing data problem nowadays. But is that solution really what you expect? That is what my dissertation is about. I want to contribute methodology that facilitates social and biomedical scientists in drawing valid inferences from incomplete data.

Why are missing data problematic?

When there are missing values in a dataset, statistical inference cannot be performed. With just one incomplete observation, simple statistics such as the mean are mathematically undefined. Statistical analyses are a means to infer conclusions about a population after observing only a subset of this population (a sample). Missing data are not a problem with purely descriptive statistics; the problems arise when we want to infer from incomplete sample to the population (or the complete sample).

Figure 2: Inference with missing data



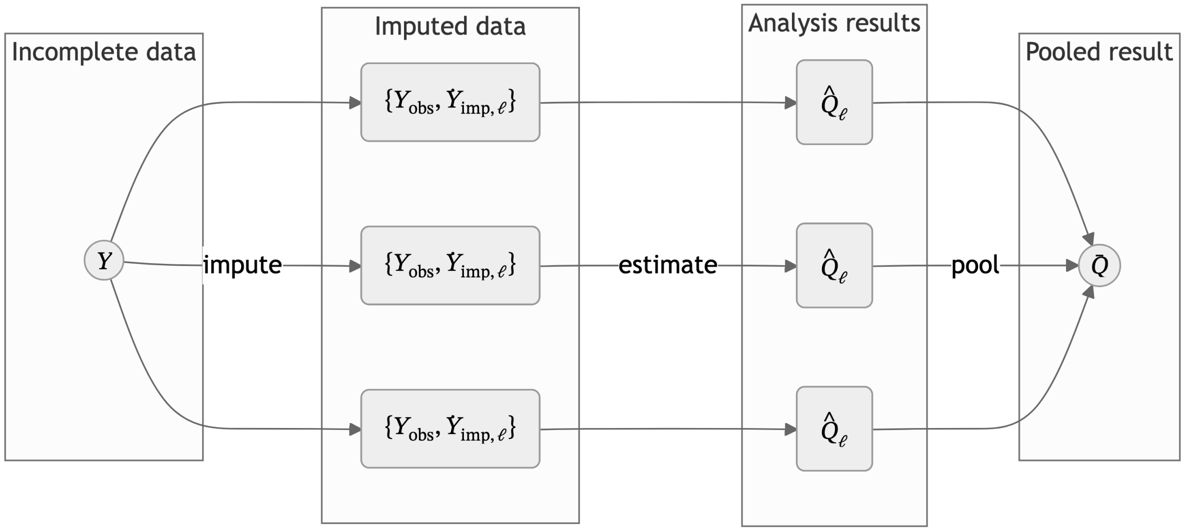
Note. Schematic view of statistical inference in the presence of non-response. Left: no missing data in the sample from the population (i.e., business as usual). Right: incomplete sample due to non-response.

To obtain results from your incomplete dataset, you will always employ a missing data strategy-–whether deliberately or not. The default strategy in most statistical software packages is to ignore all cases that have missing values (Van Buuren, 2018). This is referred to as 'list-wise deletion' (LWD). Ignoring missingness, however, lowers statistical power and may yield invalid inferences (e.g., biased estimates, spurious significant results; Van Buuren, 2018). List-wise deletion is ill-advised under most conditions (see van Buuren 2018 for exceptions). It is better to treat the missing data than to ignore it. Unfortunately, popular ad-hoc missing data methods, such as replacing missing values with variable means, lead to biased estimates. Even 'advanced' techniques for substituting missing values that recover the unobserved value almost perfectly are problematic. All techniques that rely on a single replacement value risk under-representing uncertainty in the inferences, because the uncertainty due to the non-response is not taken into account.

Multiple imputation

MI is the practice of replacing each missing value with a synthetic substitute, multiple times. For each missing datapoint, we draw samples from a distribution of values that *might have* been (posterior predictive distribution). This generates several imputations: complete*d* datasets. The completed datasets can be analyzed as if the data were complete to begin with. On each completed dataset, we obtain estimates of the analysis of scientific interest. These estimates per imputation are then combined to obtain one pooled estimate. The variance of the estimate incorporates not just the sampling variance (within each imputation) but also added variance due to non-response (between imputation variance and simulation error). The variability between imputations reflects uncertainty due to the missingness.

Figure 3: Multiple imputation workflow ($m = 3$)



Note. Read left to right. Incomplete data Y is partially observed, $Y = \{Y_{\text{obs}}, Y_{\text{mis}}\}$. The missing data Y_{mis} is imputed m times to obtain m completed datasets $\{Y_{\text{obs}}, \dot{Y}_{\text{imp},\ell}\}$, where $\ell = 1, 2, \ldots, m$. On each imputation, the quantity of scientific interest Q is estimated, yielding m analysis results $\hat{Q}_\ell$. These results are pooled across imputations to obtain a single pooled result $\bar{Q}$.

Multiple imputation means that each missing value is filled in several times, with slightly different model parameters to represent model uncertainty. The result of MI is that the uncertainty in the data due to missingness is preserved in the inference.

MI enables researchers with missing data problems to correct for differences due to missingness. Correcting for missingness happens at two levels: 1) modeling the relation between missing cases and covariates; 2) adding the appropriate variance to the models to represent the uncertainty due to missingness.

The result is a probability distribution for each missing value, from which we subsequently draw several values as imputations. If we would use infinitely many values as imputations, we would obtain the entire probability distribution. In practice, however, we use a limited number of imputations. With that, we approximate the probability distributions of missing values. The more imputations are used, the better we can represent the uncertainty in the data due to missingness.

Under the right conditions, MI yields valid, unbiased estimates (Schafer, 1997). The modeling assumptions inherent to a model-based technique like MI cannot be tested. Statistical models like MI are optimized to distinguish between effect and noise of population averages–not the individuals that make up your sample.

FCS has no (deductive) proof, only by simulation.

What is FCS?

[TODO: reframe as 'Why MI in the flavor of mice?' I.e., FCS is a flexible and agnostic way to specify conditionals to arrive at a joint. MICE also allows JOMO: a direct and inflexible way to arrive at a joint. With MICE, it's easy to generate synthetic values. mice is not just imputation software, but rather an 'information generating machine'. Describe imputation models, analysis models, non-response model and SMC.]

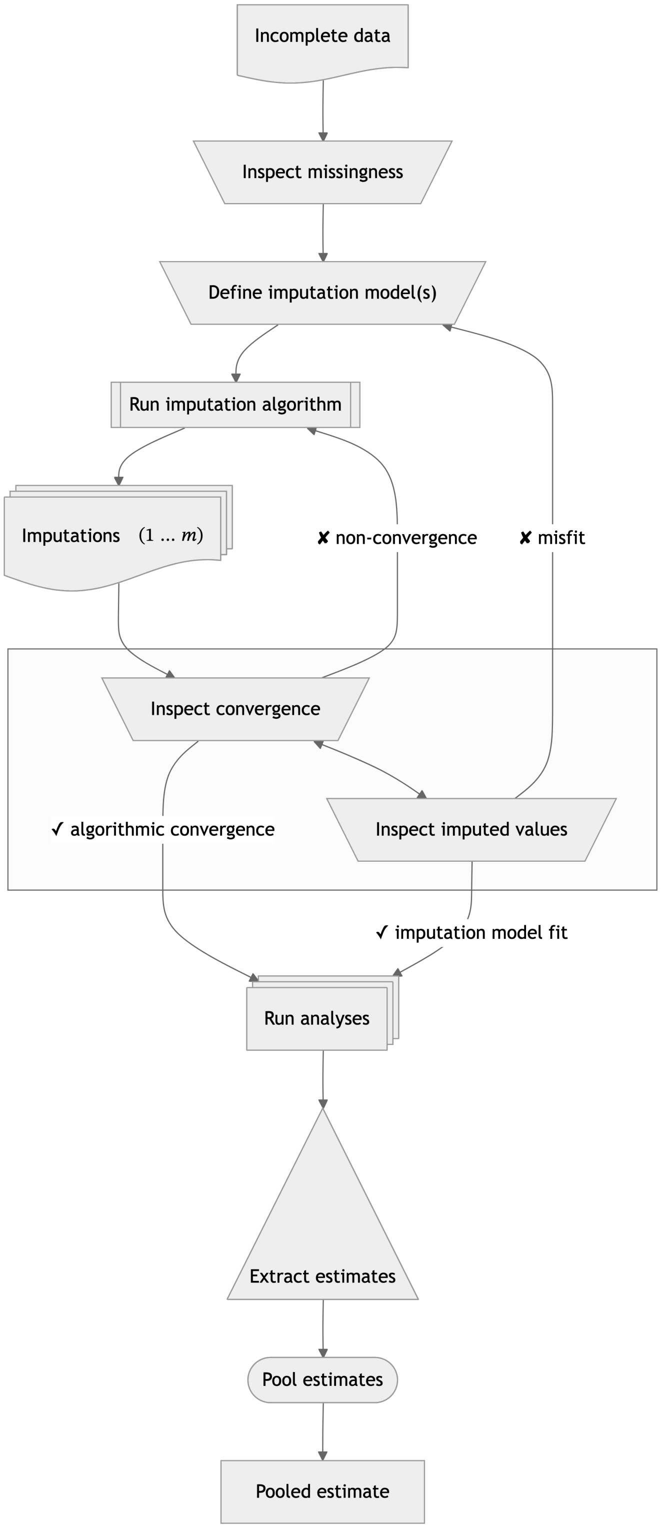
- In general, imputation methods are developed to solve missing data problems.
- What types of missing data problems are we considering in this dissertation? Incomplete samples where some—but not all—entries are missing per row or per column. Entirely missing rows would be 'unit non-response'; entirely missing columns would be unobserved variables. We are interested in data with incomplete cases/variables, or 'item non-response'.
- Missing observations in incomplete cases/variables can be filled in using an imputation model
- In the FCS framework (Van Buuren et al., 2006), we need an imputation model for each incomplete variable
- fcs is substantive model agnostic, and does not assume any specific joint distribution.
- while "Rubin (Ch. 5) distinguished three tasks for creating imputations under an explicit model: the *modeling* task the *imputation* task and the *estimation* task." (van Buuren, 2007, p. 221), FSC does not require these explicit tasks. Instead, FCS only requires an *imputation model* to describe how synthetic values may be generated, without explicit specification of the joint model.
- imputation models consist of the functional form of the model (e.g., stochastic regression) and imputation model predictors (i.e., other variables in the data)
- recurring questions: how do we choose an appropriate imputation model? and how do we know whether we have chosen an appropriate imputation model?
- "Imputation models bypass the need to specify $P(Y, X, R)$, though their use creates new responsibilities for substantiating its correctness for a given statistical analysis." (van Buuren, 2007, p. 222).
- That's where computational evaluation comes in!

FCS =/= SMC

- the quality of the imputation model can only be determined/assessed based on the complete data model –> TODO: specify terminology!! –> comp eval also possible without analysis model vs statistical evaluation relies on known analysis model (prangender vandaag want imputation gebruikt voor meer analyses) fcs is substantive model agnostic
- what if that model is not (yet) available? then take the most complex possible model and check that, because if you would take the simplest model, then that would im-

plcitly mean you are holding/constraining parameters (forcing b=0), systematically suppressing relations.

- another option would be like smc-fcs: we assume a complete data model and derive the imputations from there.


Incomplete data
Inspect missingness
Define imputation model(s)
Run imputation algorithm
Imputations (1 … m)
✘ non-convergence
✘ misfit
Inspect convergence
✓ algorithmic convergence
Inspect imputed values
✓ imputation model fit
Run analyses
Extract estimates
Pool estimates
Pooled estimate

TODO: verwerken in flow chart

- misfit en non-conv next to each other? = do the imputation models fit the data?
- Computational evaluation is in every trapezoid: processes that require human intervention. Statistical evaluation is mainly focused on the relation between the pooled estimates and
- Paragraph after this: convince reviewers that this is the right choice. Giswerk/gebrek aan controle leidt to suboptimale estimates
- missing: draw inference after pooled est; use external info after inspect miss to involva all stakeholders (CRISP-DM step)
- in simulation studies, focus is often just on incomplete data and pooled estimates

Building and evaluating imputation models

- Model building: how do we know which functional form to choose?
  - rely on defaults, tested in simulation studies
  - assess the distribution of the incomplete variable (e.g., visual inspection)
  - ...? (maybe ad transformations?)
- Model building: how do we know which imputation model predictors to select?
  - association of incomplete variable with other variables in the data ("include an auxiliary variable if its correlation with the main variable is 0.5 (in absolute value) [25]" Nguyen et al., 2021)
  - association of missingness indicator with other variables in the data
  - external input (e.g., branching patterns, or expert knowledge to correct for MNAR)
  - Take the analysis model (if available) as starting point for imputation workflow, for congeneality
  - ...?
- Model evaluation: how do we assess imputation model misfit?
  - fit of the model to the observed data
  - non-convergence in the algorithm
  - mismatch between observed and imputed data distributions (e.g., visual inspection)
  - ...?

Evaluating imputations

- FCS has no analytical proof –> rely on evidence from simulation studies
- has been shown that it works under many conditions

- but how to know in practice? we need assumptions
- like assumptions for GLM, we look at the data to evaluate the appropriateness of our chosen model
- some assumption checks are visual inspection, some are quantitative, some need external info
- similarly, we need to evaluate the choice of imputation models
- ideally: evaluate inferential validity, but only available in simulation studies
- in practice, the thing we need is not available by definition
- some assumptions are impossible to definitively determine (e.g. MNAR), some we can evaluate by looking at the data
- this is what this dissertation is about: computational evaluation
- In practice, you cannot validate whether the imputation model was appropriate (???), because what you would need to do so is exactly the thing you don't have: the missing part of the incomplete data, while you only have the observed part.
- The best thing we can do is to rely on a combination of assumptions/theoretical justifications/substantive expertise and indirect evidence of imputation model fit: check the observed versus imputed data distributions, and inspect the imputations for signs of non-convergence. This dissertation offers insight and tools to do so.
- The only setting where we do have the comparative truth (the true but missing values) is simulation studies. Imputers have to rely on simulation study results in order to choose the best imputation method for their incomplete variables. That's why simulation studies for imputation methodology should be comparable with one another. Then, imputers can make a 'grounded' assessment which imputation methods may be applicable, and choose the most appropriate one. Example: NRI search for lower bound, so impute as 'positive behavior', but per definition wrong statistical inf, because we intentionally bias in one direction
- [in simulation study kun je waarheid maken, maar in de praktijk moet je aannames doen. die aannames zijn belangrijk, juist omdat ze niet/ten dele te bewijzen zijn]
- [what is a simulation study? add a brief methodological background section on simulation studies in statistics?]

What is computational evaluation?

- Computational evaluation is not equal to statistical evaluation. Statistical evaluation is important, e.g. to quantify bias and variance of estimators/procedures, but not the focus of this dissertation.
- Computational evaluation is something to check *conditional on* acceptable statistical properties, and does not replace statistical evaluation.
- Computational evaluation is originally from information retrieval research (see e.g. "Evaluation Measures (Information Retrieval)," 2025)
- The practice of computational evaluation combines *effectiveness* (i.e., did the process yield the correct result, e.g., accuracy, precision, recall, etc.), *system quality* (e.g., speed, coverage of all possible results, etc.) and, importantly, *user utility* (e.g., happiness, productivity, etc.) (Manning et al., 2008)
- The term 'computational evaluation' is not specifically defined for the field of statistics, but it could be useful.
- One definition of computational evaluation is "to determine the quality of solutions attainable" (Dudek, 2004)
- Translated to statistics, a model may be statistically valid, but not fit for purpose. For example, computations that take too long to converge for use in a production pipeline/real-time use in patient care.
- Another definition of computational evaluation is "the process of ensuring an algorithmic solution is a good one: that it is fit for purpose" (Curzon et al., 2014).
- We can use this when a statistical process has different purposes and thus different usability/requirements. For example, calculating uncertainty at the level of a single case, a sample, or inferring to a population.
- Since the term is not defined for statistics, there is no definition to use for the computational evaluation of imputation methodology.
- In this dissertation, I want to plant a flag: how can we define 'computational evaluation' in the context of imputation methodology?
- Working definition: *systematic assessment of the algorithmic behavior and data-level outputs of imputation procedures*, focused on: algorithmic stability (convergence, failure modes), sensitivity to modeling choices, plausibility and coherence of imputed values, and usability for applied researchers.
- These aspects are often already taken into account by imputers, but I try to systematize and operationalize these aspects that are often informal or *ad hoc*.

Dimensions of computational evaluation: e.g. inferential validity, algorithmic behavior, data-level plausibility, user utility, …

- verification = "confirmation, through the provision of objective evidence, that specified requirements have been fulfilled" (ISO/IEC/IEEE) … "process of providing objective evidence that the system, software, or hardware and its associated products conform to requirements (e.g., for correctness, completeness, consistency, and accuracy)" (IEEE) …

"Verification in a life cycle context is a set of activities that compares a product of the life cycle against the required characteristics for that product. … The system has been built right. … A system could be verified to meet the stated requirements, yet be unsuitable for operation by the actual users." (IEEE)

- validation = "confirmation, through the provision of objective evidence, that the requirements for a specific intended use or application have been fulfilled" (ISO/IEC/IEEE) … "Validation in a system life cycle context is the set of activities ensuring and gaining confidence that a system is able to accomplish its intended use, goals, and objectives. The right system has been built. Validation demonstrates that the system can be used by the users for their specific tasks." (IEEE)
- verification asks whether the system has been built right whereas validation asks whether the right system has been built

Why should we care about computational evaluation of imputation methodology?

- What do we care about? bias/coverage/variance, empirical relevance/plausibility, software-engineering nitpicks like speed/convergence/fault rate.
- Computational evaluation goes beyond statistical evaluation.
- In this dissertation I will use computational evaluation as a suite of tools to investigate the domains of algorithmic stability, data plausibility, and user burden, next to inferential validity.
- Even if we know the process yields correct inferences, we can evaluate the usability/appropriateness/plausibility of the solution. For example, one question to consider is: do the imputed values lie within the range of the observed values?
- Imputation theory does not require the imputed values to be plausible.
- Rubin (1976, 1987) developed the methodology to get correct population-level inferences from incomplete samples, not at the individual case-level within the sample. [TODO: consider adding that since then, data analysis has changed]
- Bluntly stated: imputation pragmatists/purists don't care for the imputed values themselves, as long as they result in statistically valid inferences.
- But what are the limitations of purely inferential metrics (bias, coverage)? e.g. imputing negative values for age, using PMM for a variable with a sparse region (yielding 'unrealistic' imputed values), or evaluating imputed values of non-converged chain (?), or not knowing the analysis model up front so congeniality is at risk (???) -> PPC would show problem with sparse region
- And vice versa: when might the plausibility of imputed values mislead a data analyst into believing the imputations are statistically valid? when 'looking reasonable' is confused with 'being consistent with the data-generating process', such as single imputation (stoch reg), or imputing under MCAR assumption when there is a MAR mechanism, or destroying the joint distribution of the data while preserving the marginals. [TODO: figure out the difference with Gerko's interpretation of joint occurrence (cross tabs) within

conditional distribution; should we make an algorithm to show user which combinations of bivariate obs only occur in imp? Not now!]

- Statistical evaluation of bias and coverage is not possible in practice, only in simulation studies.
- Stat evalualtion may not possible IRL? Bias and confidence interval coverage are impossible to determine without a comparative truth (although PPC and bootstrapping allows us to compare with respect to observed values).
- [TODO: rephrase as ability to explain differences between observed and imputed data (e.g., lower means in boys data), not as reverse engineering but as statistical evaluation of the data –> then computational eval may be needed to explain differences?] Computational evaluation means not just taking the answer as-is, but tracing back the solution of an algorithm to its origin/back-tracing the procedure: you should *always* be able to retrieve the chain of the imputation procedure. E.g., implausible or impossible imputed values can yield valid statistical inference, but are we satisfied with the way that the imputation were obtained?

Examples of stat but not comp

- polynomials (ignoring the U in imps could still get stat val)
- example: psych study with baseline, some of which are missing; baseline, intervention, follow-up –> mice would inflate baseline and follow-up if both are incomplete; statistical evaluation will not show this (unbiased, cov ok, etc.); over-imputation or postpred check would show diff distr for obs and imp; circularity should be removed: follow-up should not predict baseline
- fcs iterations not stable, but estimand ok (in simulation study true value) –> conv at cell level??
- PMM with part of sample space with too little donors

What is the scope of this dissertation?

RQ: what tools are required to assess the quality of data-generating algorithms? in the development, evaluation, and application.

- is the status quo sufficient? no, statistical evaluation is required, but not sufficient.
- how Rubin developed the methodology works in theory, but is not enough for contemporary practitioners
- can we use existing tools or do we need more? and how? no, not all are present in the imputer's toolbox. I contribute to the new status quo ("verleggen" = generalizable scientific contribution of this dissertation)
- unifying how imputations are generated and evaluated (??)

- lifecycle of imputation methods: method is theorized, method is implemented in software, method is evaluated in simulation study, method is put to scrutiny by scientific community, method is applied in practice, method is evaluated in real-world data.
- two types of evaluation: statistical properties of the method in simulation (e.g., bias and confidence-validity; statistical evaluation) & practical application (computational evaluation).

[TODO: remove what I will not address! Maybe add a sentence 'You might expect me to go into…' In the category non-response: just item, and M(C)AR mech. In the category imputation algo: just FCS. In the category substantive models: just GLM for stat inference.] In this dissertation, we will focus on missing data problems with item non-response and MCAR/MAR missingness mechanisms. Unit non-response and MNAR missingness mechanisms are outside of the scope of this dissertation. The missing data method under consideration is primarily multiple imputation by chained equations (MICE), but some results may generalize to other iterative (FCS) imputation algorithms. We do not cover joint modeling (JOMO) imputation or non-imputation missing data methods. Although MICE is not SMC-FCS, we will restrict the analysis of scientific interest mostly to GLM family of models. The applications are aimed at statistical inference (not prediction or causal inference).

Chapters

1. Evaluation paper: evaluating (new) imputation methods via simulation studies/how we know FCS works
2. Convergence paper: evaluating imputation algorithm quantitatively
3. Software: choosing/evaluating imputation models in practice

Introducing chapter 1: evaluation

- **RQ: what to look out for when designing simulation studies for the evaluation of imputation methodology?**/How can computational evaluation criteria be systematically incorporated into simulation studies for imputation methodology?
- published as Oberman & Vink (2024), cited 27x according to Google Scholar (of which 11x CrossRef)
- based on literature review Liu et al. (2023)
- How can we trust simulation studies for imputation methods?
- How can we make them compatible & fair?

REF: Morris ADEMP: Using simulation studies to evaluate statistical methods, pre-registration template, FIMD, Introduction to statistical simulations in health research?, Pitfalls and Potentials in Simulation Studies?

Introducing chapter 2: convergence

- **RQ: FSC is iterative, so convergence *should* be a requirement for valid inference, or is it?**/How does non-convergence in iterative imputation affect inferential validity, and how can non-convergence be diagnosed?
- simulation study, published as pre-print (11x cited according to Google Scholar), presented at ICML
- even without convergence, we can get valid inference at sample/population level, but not at individual cell level
- imputation stays a custom job: convergence of imputation is conditional on imputation model
- today, imputation are not generated for 1 analysis prob, but all analyses that may be fitted on the data –> all imputation models should be compatible and congenieal –> more 'omvattende' imputation models required
- liefst wil je analysemodel parameters tracken, daarom is computational evaluation zo belangrijk
- Can we quantify the convergence of the MICE algorithm?

REFs: Graphical and numerical diagnostic tools to assess suitability of multiple imputations and imputation models, Diagnostic checking of multiple imputation models, Convergence Properties of a Sequential Regression Multiple Imputation Algorithm, FIMD, Diagnostics for multivariate imputations

Introducing chapter 3: `ggmice`

[TODO: add poster and 'age old adage' picture vs 1000w.]

This chapter introduces the software `ggmice`: an R package for building and evaluating imputation models. The `ggmice` package offers tools for the visualization of incomplete data, imputation models, and imputed data.

Indicators for adoption: how is the software already used?

- software published on Zenodo (Oberman, 2022), with open code and bug reporting via GitHub
- peer-review not formal via journal, but via Issues and Pull Requests on GitHub (10x external Issue, 13x external PR)

- impact: citations (5x according to Google Scholar), CRAN downloads (852x/month; 21k total) and conda downloads (100+/month; https://anaconda.org/channels/conda-forge/packages/r-ggmice/overview)

Design principles

- `ggmice` is compatible with the popular R packages `mice` (for imputation) and `ggplot2` (for flexible data visualizations based on the 'grammar of graphics' framework).
- `mice` imputation models are substantive model agnostic: flexible, var-by-var definition of imputation model. `ggmice` supports this.
- Grammar of Graphics = layered plots (Wilkinson, 2012) –> `ggmice` produces `ggplot` objects: layered, easily adjustable
- Assumptions: imputation model should fit the observed part, imputation model should be congenial –> providing tools to see observed and imputation side-by-side/evaluation of the distribution of observed and imputed data

Limitations

- make explicit what these plots/analyses can and cannot tell us: imputation models should fit the observed part of the data, but no way to determinately state miss mech.
- visualization =/= objective truth –> good imputation model fit/plausible values is not a guarantee for statistical validity
- visualization is not a replacement of evaluation but support, also check metrics such as FMI
- interpretation remains subjective (e.g. convergence, MAR assumption, etc.)

What does ggmice aim to solve?

The `ggmice` package aims to solve the following problems:

- no direct graphical comparisons between incomplete and imputed data; side-by-side presentation of the data before and after imputation; Visualizations with `ggmice` can be made to inspect marginal and joint distributions of incomplete variables side-to-side to compare incomplete and imputed data
- no previously existing methods for visualizing `mice` imputation models; complex imputation models are easier to review through visualization (as opposed to console prints/excel exports of predictor matrix)

- existing plotting tools for `mids` objects not easily editable/publication-ready; Visualizations with `ggmice` can be extended and adapted with tidyverse functions to create publication-ready graphs of imputed data (reflecting the uncertainty due to missingness).
- unified toolbox for the development and evaluation of imputation models

Existing solutions

The `ggmice` package fills a hiatus in the software landscape, with other packages generally supporting either `mice` or `ggplot2`, but not both. Table REF provides an overview of R packages that offer visualization tools for incomplete and imputed data. In short: with `mice` you cannot plot incomplete data distributions, all other packages lack support for `mice`-imputed data, and visualization of `mice` imputation models did not exist at all prior to `ggmice`.

	ggmice	mice	naniar	VIM
Incomplete data (e.g., `data.frame` object)				
− joint & marginal distributions	+	−	+	+
− missing data patterns	+	+	±	+
− imputation models	+	−	−	−
Imputed data (e.g., `mids` object)				
− joint & marginal distributions	+	+	−	−
− imputation models	+	−	−	−
− trace plot of the imputation algorithm	+	+	−	−
User utility (e.g., `ggplot` object)				
− flexibility/extensibility of visualization types	+	−	±	−
− adaptability/publication readiness of visualizations	+	−	+	−

TODO

- write table footnote to explain all aspects of the table. i.e., these are only diffs. the table does not include things that all packages can –> maybe as appendix?
- explain the purpose of the table: I'm both player and referee. I've got skin in the game, but I made the rubric. Someone else might evaluate/value different aspects.
- also add number of downloads: `ggmice` has 800 monthly compared to `VIM` 13k and `naniar` 22k
- make 'Imputation models' a separate section?

Software development

- Open Source Software development/FAIR –> R, GitHub, CRAN, Zenodo, …

- CRAN downloads
- GitHub stars and issues
- StackOverflow mentions/questions
- feedback at conferences
- citations of the package (Zenodo reference)

Future research & development

- visualizing `mira` objects
- add posterior predictive check plots & over-imputation
- quantifying uncertainty at pattern/cell level
- tools for sensitivity analyses: inducing MNAR in observed part of the data and evaluating effects on the imputations/inference
- What data-level diagnostics are informative for assessing imputation model adequacy in practice (i.e., when inferential validity cannot be evaluated)?
- pre vs post imputation facilitator functies om te checken of er nieuwe verbanden in zitten (kruistabel)

Chapter 1: evaluation

[TODO: link/insert]

Chapter 2: convergence

[TODO: link/insert]

Chapter 3: ggmice

[TODO: link/insert]

Discussion

Conclusion

- Computational evaluation of imputation methodology is never finished.
- The main findings in this thesis are…

  - Defining and testing new imputation methods through simulation studies requires careful consideration of the simulation design and evaluation to make imputation methods comparable across studies. We need simulation studies to know imputation methods work. But we cannot be certain they actually *did* work whenever we apply them. The missing part of the data is by definition not available, so we cannot definitively state anything about the imputation model fit to the unobserved part of the data. Moreover, there's untestable assumptions (e.g., MNAR) and lack of clear diagnostic cut offs (e.g., convergence).
  - Non-convergence is hard to diagnose, and typical thresholds to evaluate non-convergence may not be applicable to FCS algorithms: MICE reaches stable output before non-convergence metrics would indicate so. There is a mismatch between theoretical convergence criteria and practical algorithmic behavior. That's why we still have to rely on visual inspection.
  - Visual inspection aids imputers in developing and evaluating imputation models, because formal tests are not available. The R package `ggmice` offers tools for the computational evaluation of the imputations models that were previously unavailable.

- Recommendations for fellow missing data methodologists:

  - design simulation studies structured and reproducible (e.g., report design choices, publish code)
  - talk to your intended audience (i.e., applied researchers), or at least check what they are struggling with (e.g., StackOverflow)
  - …?

- Recommendations for fellow imputers:

  - think before you code
  - look at the data, plot the data
  - …?

Limitations

- Computational evaluation is not a substitute for thinking. It may give a false sense of certainty, e.g. against MNAR mechanism (untestable assumption).

- Instead, use computational evaluation as complement to (not a replacement for) substantive knowledge and sensitivity analyses. Use expert insight
- This entire dissertation relies on the R language and centers the `mice` package [TODO rephrase as not a flaw: I focus on a dominant ecosystem, and concepts generalize beyond].
  - While this set-up is very popular, the data science landscape has expanded and is ever evolving
  - Many machine learning and deep learning methods are not (yet) implemented as imputation models in `mice`. These might be able to better approximate the posterior predictive distribution of the missing values than `mice` methods, but research should show that still.
- ~~I haven't tested whether the tools I developed actually improve the imputations of `mice` users~~ This dissertation evaluates tools and diagnostics at the methodological level; assessing their causal impact on user behavior and imputation quality is not feasible to systematically test.
  - GitHub issues and StackOverflow are indication
  - Onmogelijk om evidence-based conclusies te trekken, maar misschien wel evidence-informed?
- The visualizations implemented in `ggmice` are generally better suited for numeric data, as opposed to categorical variables (or even open text field data, which is not supported by `mice` currently). Future research on: associations of cat vars with miss indicators, and convergence parameter other than SD of imputed values.

> - TODO: rephrase 'haven't's into future research (positive phrasing: reframe as scope conditions, not shortcomings)
> - maybe future: computational evaluation of variable importance within imputation models

Outlook

- This chapter ends with a future perspective…
- Missing data remains a problem indefinitely. I expect imputation methodology to be developed and applied for the foreseeable future. But the way we deal with data might change.
- With the rise of machine learning and deep learning methods (or "AI"), evaluation will become ever more important. Novel methods are being developed under a prediction framework, not statistical inference framework: ignoring the uncertainty in the estimates,

and thus not incorporating between-imputation variance. This leads to too narrow CIs, and too low p-values. Which, in turn, causes spurious results.

- AI models should propagate uncertainty!
- trust in science requires honest reflection of uncertainty in our analyses
- be open about uncertainty, and limitations of scientific studies
- publish null results (e.g., registered reports or pre-registrations)
- share data and code with other scientists, and the public
- publish open access, educational materials too!
- change the way we evaluate science: this dissertation is an example of the new recognition and rewards vision. it includes software as an actual chapter, not something extra.
- I hope to inspire others to reflect on their own view of what scientific output is. And most of all, I hope to set an example for other PhD candidates to do the same
- imputation is not always the way to go: missingness can tell you about mismatch between data collection method and lived experience –> positionality??

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